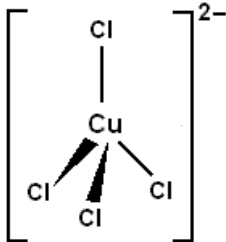


| Question |     |  |  | Marking details                                                                                                                                                                                                             | Marks available |     |     |       |       |      |
|----------|-----|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |  |  |                                                                                                                                                                                                                             | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 7        | (a) |  |  | $\text{Fe}_3\text{O}_4 + 4\text{CO} \rightarrow 3\text{Fe} + 4\text{CO}_2$<br><br>ignore state symbols                                                                                                                      |                 | 1   |     | 1     |       |      |
|          | (b) |  |  | award (1) for either of following <ul style="list-style-type: none"><li>stability of +2 oxidation state increases down the group (due to the inert pair effect)</li><li>carbon stable as +4 and lead stable as +2</li></ul> | 1               |     |     | 1     |       |      |
| 8        |     |  |  |                                                                                                                                            |                 | 1   |     | 1     |       | 1    |
|          |     |  |  | Section A total                                                                                                                                                                                                             | 4               | 6   | 0   | 10    | 0     | 1    |